

## FET I EIDFJORD, Norway

WMO: 013400

Lat: 60.4083N

Lon: 7.2797E

Elev: 2418

StdP: 13.46

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 1.2        | 4.2 | -7.5                          | 4.0 | 4.2  | -4.0 | 4.9 | 4.6  | 32.4                  | 19.4 | 30.1 | 20.4 | 7.7                   | 130  | 0.746 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.0                   | 71.5            | 56.3 | 67.8 | 55.4 | 64.2 | 53.9 | 58.4                | 67.7 | 56.7 | 65.1 | 55.1 | 62.4 | 6.9                  | 280  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 54.4                            | 68.9 | 61.2 | 52.8 | 64.9 | 60.1 | 51.3 | 61.3 | 58.4 | 26.3          | 67.4 | 25.2 | 65.4 | 24.1 | 62.5 | 63.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 26.6          | 23.0 | 19.8 | DB                         | -4.5 | 74.5               | 2.8 | 2.8                                                | -6.5 | 76.5       | -8.1 | 78.1       | -9.7  | 79.7       | -11.8 | 81.7 | (4) |
| (5)               |      |      | WB                         | -4.7 | 60.6               | 3.0 | 2.3                                                | -6.9 | 62.2       | -8.6 | 63.6       | -10.3 | 64.8       | -12.5 | 66.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 37.0   | 22.8  | 23.0  | 26.4  | 33.9  | 42.3  | 48.8  | 53.6  | 51.8  | 46.4  | 38.0  | 30.2  | 25.6 |
|                                                                                  | DBStd       | 13.03  | 8.31  | 8.47  | 8.32  | 6.41  | 6.68  | 5.53  | 6.04  | 4.65  | 5.52  | 6.34  | 7.78  | 8.68 |
|                                                                                  | HDD50       | 5082   | 844   | 757   | 733   | 482   | 256   | 86    | 29    | 35    | 136   | 374   | 593   | 757  |
|                                                                                  | HDD65       | 10231  | 1309  | 1177  | 1198  | 932   | 704   | 486   | 355   | 410   | 558   | 837   | 1043  | 1222 |
|                                                                                  | CDD50       | 328    | 0     | 0     | 0     | 0     | 17    | 50    | 142   | 90    | 27    | 2     | 0     | 0    |
|                                                                                  | CDD65       | 3      | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 34     | 0     | 0     | 0     | 0     | 2     | 1     | 30    | 1     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 8.4    | 10.6  | 9.6   | 9.3   | 8.3   | 7.6   | 6.9   | 6.4   | 6.4   | 7.4   | 8.6   | 9.6   | 9.6  |
| Precipitation                                                                    | PrecAvg     | 43.4   | 4.7   | 4.1   | 4.0   | 2.2   | 2.0   | 2.5   | 3.3   | 3.4   | 4.1   | 4.2   | 4.1   | 4.9  |
|                                                                                  | PrecMax     | 60.4   | 10.2  | 9.4   | 16.5  | 4.9   | 4.1   | 5.5   | 6.1   | 6.1   | 7.8   | 8.5   | 11.9  | 12.6 |
|                                                                                  | PrecMin     | 23.5   | 0.0   | 0.0   | 0.0   | 0.5   | 0.4   | 0.6   | 1.5   | 1.3   | 1.2   | 1.0   | 0.0   | 0.0  |
|                                                                                  | PrecStd     | 8.9    | 3.0   | 2.8   | 3.3   | 1.2   | 0.9   | 1.3   | 1.1   | 1.2   | 1.8   | 2.0   | 2.8   | 3.6  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 40.0  | 41.3  | 44.2  | 53.3  | 69.7  | 72.6  | 77.4  | 71.2  | 66.5  | 57.7  | 48.7  | 42.6 |
|                                                                                  |             | MCWB   | 36.8  | 37.2  | 37.2  | 42.7  | 52.8  | 54.7  | 59.3  | 56.9  | 53.7  | 51.2  | 44.4  | 39.3 |
|                                                                                  | 2%          | DB     | 37.8  | 37.5  | 41.0  | 48.8  | 62.9  | 69.0  | 73.5  | 66.9  | 61.1  | 52.4  | 44.9  | 39.9 |
|                                                                                  |             | MCWB   | 35.3  | 34.9  | 36.2  | 41.1  | 50.4  | 53.6  | 58.4  | 55.5  | 52.5  | 46.1  | 41.2  | 37.3 |
|                                                                                  | 5%          | DB     | 35.7  | 35.4  | 38.4  | 45.5  | 58.2  | 64.2  | 69.7  | 63.5  | 58.1  | 49.8  | 42.4  | 37.6 |
|                                                                                  |             | MCWB   | 33.3  | 33.0  | 35.0  | 39.4  | 47.5  | 51.2  | 57.2  | 54.4  | 51.3  | 44.9  | 39.4  | 35.2 |
|                                                                                  | 10%         | DB     | 33.6  | 33.4  | 36.3  | 42.9  | 53.5  | 60.2  | 65.5  | 60.5  | 55.1  | 47.1  | 40.0  | 35.7 |
|                                                                                  |             | MCWB   | 31.4  | 31.0  | 33.3  | 37.9  | 45.2  | 49.7  | 55.9  | 53.0  | 49.9  | 43.0  | 37.2  | 33.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 37.6  | 37.2  | 38.7  | 44.0  | 54.3  | 56.6  | 61.6  | 58.5  | 56.1  | 51.4  | 44.7  | 40.0 |
|                                                                                  |             | MCDB   | 39.3  | 39.5  | 42.4  | 50.4  | 66.0  | 67.5  | 73.1  | 66.7  | 63.4  | 56.9  | 48.2  | 42.0 |
|                                                                                  | 2%          | WB     | 35.7  | 35.6  | 37.1  | 41.7  | 51.2  | 54.6  | 59.8  | 57.0  | 53.7  | 47.5  | 41.8  | 37.7 |
|                                                                                  |             | MCDB   | 37.6  | 37.3  | 39.7  | 47.4  | 61.5  | 65.8  | 70.1  | 64.4  | 59.3  | 51.2  | 44.5  | 39.6 |
|                                                                                  | 5%          | WB     | 33.5  | 33.4  | 35.5  | 40.0  | 48.3  | 52.7  | 58.0  | 55.5  | 51.9  | 45.4  | 39.7  | 35.5 |
|                                                                                  |             | MCDB   | 35.3  | 34.9  | 38.2  | 45.1  | 56.5  | 62.3  | 67.4  | 61.7  | 56.9  | 49.1  | 42.3  | 37.1 |
|                                                                                  | 10%         | WB     | 31.6  | 31.4  | 33.3  | 38.3  | 45.7  | 50.8  | 56.3  | 53.9  | 50.0  | 43.3  | 37.4  | 33.7 |
|                                                                                  |             | MCDB   | 33.3  | 33.1  | 36.0  | 42.6  | 52.7  | 58.5  | 64.6  | 59.4  | 54.9  | 46.5  | 39.7  | 35.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.0   | 9.6   | 11.0  | 12.3  | 13.9  | 15.9  | 15.0  | 14.0  | 11.9  | 10.5  | 8.6   | 9.0  |
|                                                                                  |             | MCDBR  | 9.0   | 9.4   | 11.5  | 14.3  | 20.9  | 23.4  | 22.3  | 18.6  | 16.8  | 14.3  | 10.5  | 9.5  |
|                                                                                  |             | MCWBR  | 8.0   | 7.4   | 8.1   | 8.8   | 11.4  | 11.9  | 10.2  | 9.8   | 9.8   | 9.2   | 8.4   | 8.3  |
|                                                                                  | 5% WB       | MCDBR  | 9.2   | 9.0   | 10.6  | 13.6  | 19.4  | 19.9  | 20.0  | 16.9  | 14.7  | 11.8  | 10.5  | 9.3  |
|                                                                                  |             | MCWBR  | 8.3   | 7.8   | 7.7   | 8.5   | 10.9  | 11.1  | 10.0  | 9.8   | 9.2   | 8.7   | 8.6   | 8.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.217  | 0.257 | 0.291 | 0.324 | 0.330 | 0.327 | 0.341 | 0.333 | 0.304 | 0.283 | 0.227 | 0.189 |      |
|                                                                                  | taud        | 2.124  | 2.237 | 2.192 | 2.163 | 2.269 | 2.429 | 2.424 | 2.439 | 2.509 | 2.415 | 2.158 | 2.001 |      |
|                                                                                  | Ebn at Noon | 181    | 239   | 265   | 276   | 283   | 287   | 279   | 272   | 264   | 229   | 172   | 138   |      |
|                                                                                  | Edh at Noon | 12     | 21    | 30    | 38    | 37    | 33    | 32    | 29    | 23    | 18    | 12    | 9     |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 83     | 250   | 658   | 1244  | 1518  | 1522  | 1452  | 1134  | 719   | 334   | 114   | 55    |      |
|                                                                                  | RadStd      | 21     | 70    | 129   | 214   | 264   | 182   | 183   | 101   | 83    | 49    | 18    | 7     |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (50 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page